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# Activity - 8

Roll no: 19

Name: K. Akhileswara Reddy

Course code: DSA061

Regno: 192385068

Series SP-C19: clinic visits

Task 1: correlation matrix

| Variable | pressure | coffee | late_entries | room_dev | print_jobs |
|---|---|---|---|---|---|
| pressure | 1.00 | 0.88 | -0.92 | 0.26 | 0.43 |
| coffee | 0.88 | 1.00 | -0.85 | 0.36 | 0.20 |
| late_entries | -0.92 | -0.85 | 1.00 | -0.32 | 0.01 |
| room_dev | 0.26 | 0.36 | -0.32 | 1.00 | -0.07 |
| print_jobs | -0.23 | -0.26 | 0.26 | -0.33 | 0.02 |

- n = 15
- Distinct pairs = 21

Task 2: Strongest clues

Strongest positive: Pressure and coffee, $r \approx 0.88$, strong.

Strongest negative: Pressure and late_entries $r \approx -0.92$, strong.

Task 3: The deceptive Zero.

- Pearson correlation: $r \approx -0.03$.
- Quadratic fit: $R^2 \approx 0.98$.

Task 4: The outlier alibi

- Suspicious row: 6.
- Correlation with all rows: $r \approx 0.69$
- correlation without row 6: $r \approx 0.28$.

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Task 5: The prime suspect

- r(coffee, late_entries) ≈ −0.81,
- r(coffee, pressure) ≈ 0.88.
- r(pressure, late_entries) ≈ −0.92
- Partial correlation controlling for pressure r ≈ −0.20.

Verdict: coffee sales do not establish that coffee causes late-night entries; exam pressure may influence both variables.